**FYUGP B.Sc. MATHEMATICS HONOURS**

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

**MODULE 3: PERMUTATIONS, ORBITS, EVEN AND ODD PERMUTATIONS, COSETS AND LAGRANGE**

Past questions

## Section A: 1 to 3 marks

***Permutations, products and disjoint cycles***

1 Write the order of the permutation $(1,2)(1\,9\,8)$ in $S_9$. **(1 mark)** *November 2018*

2 Compute $(1,4)(7,5)(2,5,7)$ in $S_7$. **(1 mark)** *November 2018*

3 The smallest non abelian group has _____ number of elements. **(1 mark)** *November 2021*

4 Find the order of the permutation $(1,3)(2,6)(1,4,5)$. **(3 marks)** *November 2021*

5 Consider the following permutations in $S_7$ : $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 2 & 5 & 4 & 6 & 1 & 7 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 1 & 5 & 7 & 4 & 6 & 3 \end{pmatrix}$. Compute $\sigma\tau$ and $\tau\sigma$. **(2 marks)** *November 2022*

6 Find the order of the permutation $(1,2,5)\,(2,3,4)\,(5,6)$. **(2 marks)** *November 2022*

7 Find the order of the permutation $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 5 & 3 & 2 & 1 \end{pmatrix}$. **(2 marks)** *November 2024*

8 Write the permutation: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 2 & 7 & 6 & 3 & 8 & 1 & 4 \end{pmatrix}$ in $S_8$ as disjoint product of cycles and as a product of transpositions. **(2 marks)** *November 2025*

9 Find the order of the permutation $(1,2,3)\,(2,4,3,5)\,(1,3,2)$ in $S_5$. **(2 marks)** *November 2025*

***Orbits***

10 What are the orbits of the identity permutation $\sigma$ of a set $A$ ? **(1 mark)** *November 2021*

***Even and odd permutations and the alternating group***

11 Show that the set of all even permutations of $S_n$ is a sub-group of $S_n$. **(2 marks)** *November 2025*

***Cosets and index***

12 What is the index of $A_n$ in $S_n$. **(1 mark)** *November 2018*

13 Let $G = \mathbb{Z}_{12}$, and let $H$ be the subgroup $4\mathbb{Z}_{12}$. Find all cosets of $H$. **(2 marks)** *November 2022*

14 Let $H$ be a subgroup of the group $G$. For $a, b \in G$, define $a \sim b$ if $ab^{-1} \in H$. Prove that $\sim$ is an equivalence relation. **(2 marks)** *November 2023*

***The theorem of Lagrange***

15 State True or False. “Every group of order 31 is cyclic”. **(1 mark)** *November 2021*

16 Prove that any group of prime order is cyclic. **(2 marks)** *November 2024*

## Section B: 4 to 7 marks

***Permutations, products and disjoint cycles***

17 Show that every permutation $\sigma$ of a finite set is a product of disjoint cycles. **(7 marks)** *November 2018*

18 What is the order of the cycle $(1,4,5,7)$ in $S_8$ ? **(4 marks)** *November 2021*

19 (a) Let $\sigma \in S_n$ be written as a product of disjoint cycles, prove that the order of $\sigma$ is the least common multiple of the lengths of its cycles. (b) Find the order of $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 5 & 3 & 2 & 1 \end{pmatrix}$. **(5 marks)** *November 2023*

20 Write $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 4 & 10 & 5 & 7 & 8 & 2 & 6 & 9 & 1 \end{pmatrix}$ as product of disjoint cycles. Find its order and inverse. **(5 marks)** *November 2023*

21 Prove that every permutation in $S_n$ can be written as a product of disjoint cycles. **(5 marks)** *November 2025*

***Even and odd permutations and the alternating group***

22 Show that the permutation $(1,4,5,6)\,(2,3,1,5)$ is an even permutation. **(4 marks)** *November 2021*

23 Prove that set of all even permutations of $S_n$ is a subgroup of $S_n$. **(6 marks)** *November 2021*

***Cosets and index***

24 Write all the left cosets of $3\mathbb{Z}$ in $\mathbb{Z}$. **(4 marks)** *November 2018*

25 Find the partition of the group $\mathbb{Z}_6$, under the operation addition modulo 6, into cosets of the subgroup $H = \{0, 3\}$. **(4 marks)** *November 2021*

26 Let $G = \mathbb{Z}_{24}$ and $H = \langle [3] \rangle$. Find all cosets of $H$. **(5 marks)** *November 2023*

***The theorem of Lagrange***

27 Show that every group of prime order is abelian. **(4 marks)** *November 2018*

28 Define order of an element in any group $G$. Show that in a finite group $G$, order of any element divides order of $G$. **(7 marks)** *November 2018*

29 Show that every proper subgroup of a group $G$, with $o(G) = pq$ where $p$ and $q$ are prime, is cyclic. **(7 marks)** *November 2018*

## Section C: 10 marks and above

***Permutations, products and disjoint cycles***

30 (a) Define the term orbit, cycle and transposition with respect to a permutation. (b) Write the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix}$ as product of disjoint cycles. (c) Define even and odd permutation. Write $(1,4,5,6)(2,1,5)$ as a product of transpositions. **(13 marks)** *November 2021*

31 (a) Prove that every permutation in $S_n$ can be written as a product of disjoint cycles. (b) Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 2 & 7 & 6 & 3 & 8 & 1 & 4 \end{pmatrix}$ be a permutation in $S_8$. Express $\sigma$ as a product of disjoint cycles. **(10 marks)** *November 2024*

32 Let $S$ be any set and $\sigma, \tau$ are disjoint cycles in $\mathrm{Sym}(S)$. Show that $\sigma\tau = \tau\sigma$. **(10 marks)** *November 2025*

***Even and odd permutations and the alternating group***

33 If permutation written as a product of transpositions in two ways, prove that the number of transpositions is either even in both cases or odd in both cases. **(10 marks)** *November 2023*

***The theorem of Lagrange***

34 (a) State and prove Lagrange theorem. (b) Prove that any group of prime order is cyclic. **(10 marks)** *November 2023*

35 Let $H$ be a sub-group of the finite group $G$. Show that the order of $H$ is a divisor of order of $G$. **(10 marks)** *November 2025*